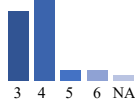


An undirected graph with edge list: $[[0, 1], [1, 2], [2, 3], [3, 4], [0, 4], [2, 4], [0, 3], [2, 5], [5, 6], [6, 7], [7, 8], [5, 8], [5, 7], [6, 8]]$. What is the diameter of this graph? Only provide a concrete number.

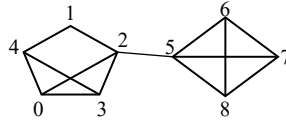
Predicted diameter distribution



GPT-4-turbo

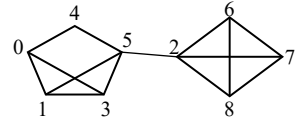
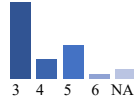


Mistral-7B



An undirected graph with edge list: $[[4, 1], [1, 2], [2, 3], [3, 0], [4, 0], [2, 0], [4, 3], [2, 5], [5, 6], [6, 7], [7, 8], [5, 8], [5, 7], [6, 8]]$. What is the diameter of this graph? Only provide a concrete number.

Predicted diameter distribution



An undirected graph with edge list: $[[0, 4], [4, 5], [5, 3], [3, 1], [0, 1], [5, 1], [0, 3], [5, 2], [2, 6], [6, 7], [7, 8], [2, 8], [2, 7], [6, 8]]$. What is the diameter of this graph? Only provide a concrete number.

Predicted diameter distribution

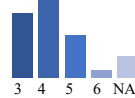


Figure 5: Illustrative Examples of LLMs Failure to Solve Graph Computational Problems under Permutation (i.e., node re-ordering). Experiments were conducted for **30** times.

of one-step DP update such that no log-precision constant-width constant-depth transformer can simulate.

Proof. We present a proof by contradiction. Assume that a token in a Transformer with a constant depth, constant width, and log-precision can attend to only a constant number of nodes. Under this